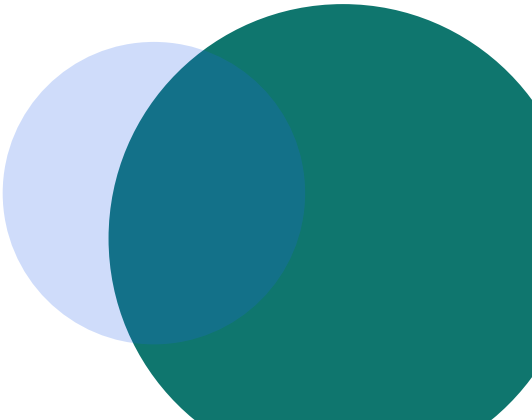
Purse Autonomous Finance

Product Requirements Document

Product vision, requirements, personas, and success criteria

Version	2.4.0
Product Lead	Autonomous Finance Core Team
Status	Approved & Implemented
Reference repository	`dwmk/purse.ai`



Typeset edition of the supplied PRD Markdown.

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1 Vision and Executive Summary

Purse is an edge-native, decentralized personal finance copilot designed to free individuals from opaque cloud databases and cumbersome financial spreadsheets. Built around **Fiona**, an embodied 3D virtual financial secretary, Purse combines conversational intelligence, local-first zero-knowledge data security, and multi-asset bookkeeping.

Whether a user is unbanked, relies primarily on physical cash, manages Web3 crypto assets across multiple blockchains, or operates traditional checking and savings accounts, Purse aims to provide a unified, private command center.

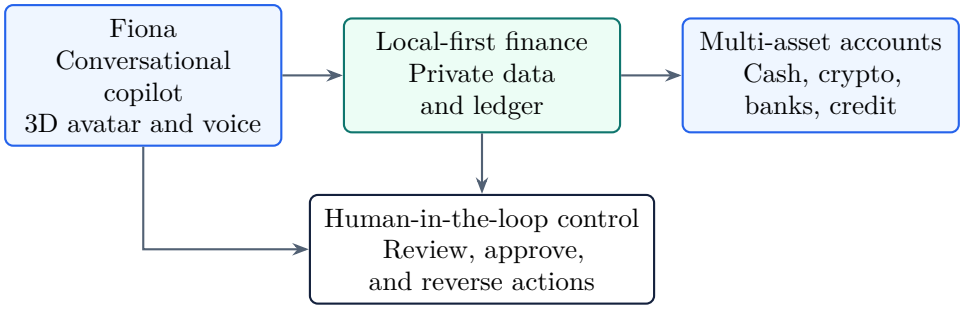


Figure 1: Product pillars represented in the PRD.

2 Core Problem Statement

The product is intended to address four problems:

1. **Privacy concerns:** Mainstream personal finance aggregators are described as storing sensitive transaction logs on remote servers, monetizing spending patterns, and requiring persistent OAuth connections.
2. **Exclusion of cash and crypto:** Many tools assume financial activity occurs through domestic checking accounts or credit cards, leaving physical cash and self-custody Web3 assets poorly supported.
3. **Rigid AI chatbots:** Existing assistants may struggle to answer general questions without redirecting the conversation to account balances or returning canned responses.
4. **Risky autonomous write access:** Users may worry about AI hallucinations creating transactions or modifying accounts without inspection and approval.

3 Product Features and Requirements

3.1 Conversational Copilot and Intelligent Persona: Fiona

3.1.1 General knowledge queries

Fiona must answer general questions conversationally, accurately, and thoroughly using the currently selected LLM. Example topics in the source include BRAC University, dollar-cost averaging, and inflation. Fiona must not deflect from general inquiries or force the conversation back to bank balances when the user asks an external question.

3.1.2 Autonomous financial operations

The model evaluates whether an instruction requires a financial operation, such as adding transactions, creating accounts, scheduling bills, or recording memory rules. When a financial modification is required, Fiona formulates a structured `action_json` proposal attached to her response.

3.1.3 Supported LLM selection

Table 1: LLM options listed in the PRD.

Provider / mode	Model	Role / notes
Google Gemini	`gemini-3.8-flash`	Default model
OpenAI	`gpt-4o-mini`	Hosted model option
DeepSeek	`deepseek-chat`	Hosted model option
Zhipu	`glm-4-flash`	Hosted model option
Browser Edge LM	`smolLM2-135M-instruct`	Offline browser inference option

3.1.4 3D interactive avatar and voice

- Real-time 3D VRM avatar with Mixamo-retargeted idle animations.
- Phoneme-driven procedural lip-sync.
- Web Speech API integration for real-time speech-to-text (STT) and text-to-speech (TTS) narration.

3.2 Multi-Asset Account Topology

Table 2: Account classes and product requirements.

Account class	Requirements	Purpose / examples
Cash (default)	Create a Physical Cash Wallet without requiring a bank or institution; include cash in liquidity, runway, and safe daily spend calculations.	Pocket cash, cash-first users, petty cash.
Crypto wallets	Support multi-chain networks including Solana, Ethereum, Bitcoin, and Polygon; store public wallet address and network identifier; show crypto badges.	Self-custody and Web3 assets.
Bank and custodial accounts	Support checking, high-yield savings vaults, credit cards, and investment brokerages; institution names and masked account numbers such as •••• `1234`.	Traditional financial accounts.
Inter-account transfers	Transfer between accounts and update source and destination balances atomically.	ATM cash withdrawals or fiat-to-crypto transfers.

3.3 Safe Execution Proposal Gate

- **Human-in-the-loop safeguard:** Every mutating action generated by Fiona (transactions, accounts, bills, or rules) must appear as an interactive Execution Gate Card in chat.
- **Pending state:** Actions remain pending until the user selects **Approve & Execute**.
- **One-click reversibility:** Every committed action can be rolled back from the Overview dashboard or action log, reverting balance changes and ledger entries.

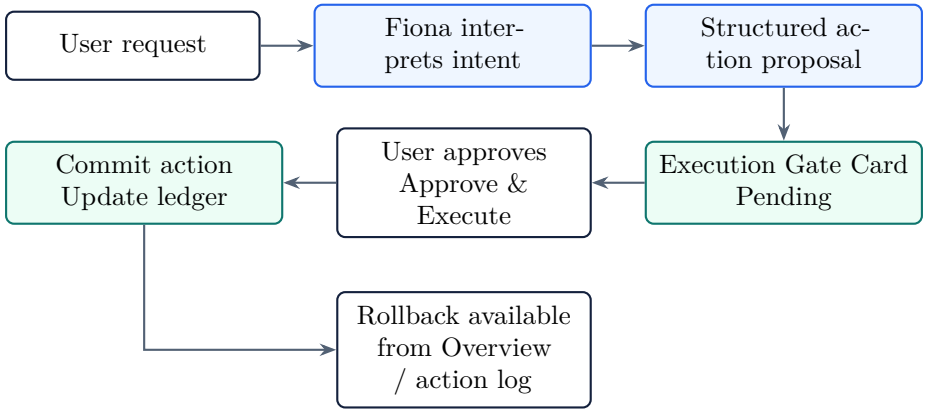


Figure 2: Human approval and rollback lifecycle for financial mutations.

3.4 Data Ingestion and Statement Parser

- **AI-powered extraction:** Accept CSV exports, bank statements, or unstructured text notes, and organize them into accounts, categorized transactions, and upcoming bills.
- **Deterministic local parser:** Provide a fallback regular-expression parser that operates entirely client-side.
- **Clean scratch initialization:** Offer a quick-start path that creates a default Cash account without dependencies.

4 User Personas

4.1 Persona 1: Tariq — Privacy-Conscious Cash and Crypto Earner

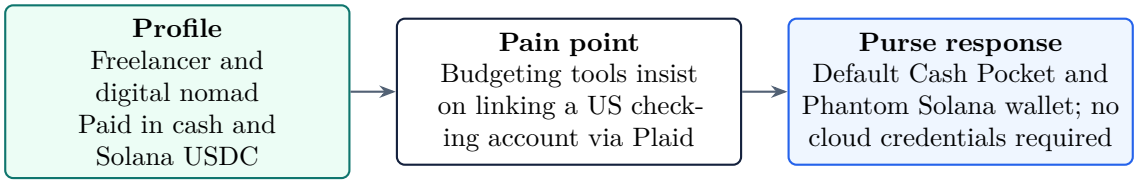


Figure 3: Tariq's need and the intended product response.

4.2 Persona 2: Elena — Multi-Account Optimizer

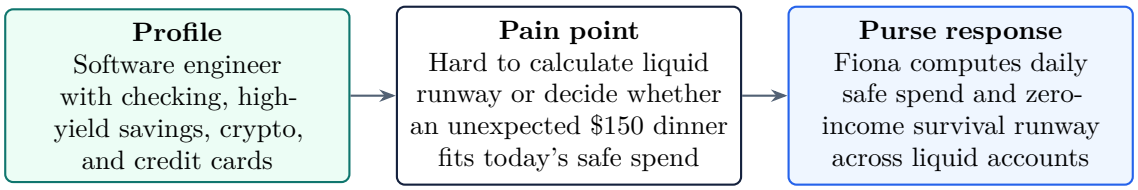


Figure 4: Elena's need and the intended product response.

5 Non-Functional Requirements

Table 3: Non-functional requirements (NFRs).

Category	Target	Requirement
Performance	Under 10 ms	In-browser vector search completes within 10 ms; streamed AI chunks render with zero layout shift.
Availability / offline resilience	Offline mode	Browser Edge LM operates through WebGPU / CacheStorage without internet connectivity.
Data security	Zero telemetry	No external user-tracking analytics and no server-side transaction logging.
Compatibility	Responsive / PWA	Support mobile, tablet, and widescreen desktop displays; comply with PWA expectations.

6 Success Metrics and KPIs

Table 4: Success metrics specified by the PRD.

Metric	Target	Definition in source
Cognitive accuracy	$> 95\%$	Accuracy in distinguishing general knowledge queries from ledger mutations.
Execution safety	100%	All LLM-initiated financial mutations pass through user confirmation.
Time to first record	Under 30 seconds	New users initialize a clean Cash or Crypto ledger in less than 30 seconds.

Document Notes

This LaTeX edition reorganizes the supplied PRD into a typeset document with tables and diagrams. Product requirements, model names, personas, and targets are based on the source document; no external verification or expansion has been added.